

Chapter 4 : Intermediate SQL

Outline

- Authorization

Authorization

- We may assign a user several forms of authorizations on parts of the database.
 - **Read** - allows reading, but not modification of data.
 - **Insert** - allows insertion of new data, but not modification of existing data.
 - **Update** - allows modification, but not deletion of data.
 - **Delete** - allows deletion of data.
- Each of these types of authorizations is called a **privilege**. We may authorize the user all, none, or a combination of these types of privileges on specified parts of a database, such as a relation or a view.

Authorization (Cont.)

- Forms of authorization to modify the database schema
 - **Index** - allows creation and deletion of indices.
 - **Resources** - allows creation of new relations.
 - **Alteration** - allows addition or deletion of attributes in a relation.
 - **Drop** - allows deletion of relations.

Authorization Specification in SQL

- The **grant** statement is used to confer authorization
 - grant** <privilege list>
 - on** <relation or view >
 - to** <user list>
- <user list> is:
 - a user-id
 - **public**, which allows all valid users the privilege granted
 - A role (more on this later)
- Example:
 - **grant select on department to** Amit, Satoshi
- Granting a privilege on a view does not imply granting any privileges on the underlying relations.
- The grantor of the privilege must already hold the privilege on the specified item (or be the database administrator).

Privileges in SQL

- **select**: allows read access to relation, or the ability to query using the view
 - Example: grant users U_1 , U_2 , and U_3 **select** authorization on the *instructor* relation:

grant select on *instructor* to U_1 , U_2 , U_3

- **insert**: the ability to insert tuples
- **update**: the ability to update using the SQL update statement
- **delete**: the ability to delete tuples.
- **all privileges**: used as a short form for all the allowable privileges

grant update (budget) on department to Amit, Satoshi;

Revoking Authorization in SQL

- The **revoke** statement is used to revoke authorization.
revoke <privilege list> **on** <relation or view> **from** <user list>
- Example:
revoke select on student from U_1, U_2, U_3
- <privilege-list> may be **all** to revoke all privileges the revokee may hold.
- If <revokee-list> includes **public**, all users lose the privilege except those granted it explicitly.
- If the same privilege was granted twice to the same user by different grantees, the user may retain the privilege after the revocation.
- All privileges that depend on the privilege being revoked are also revoked.

Roles

- A **role** is a way to distinguish among various users as far as what these users can access/update in the database.
- To create a role we use:
create a role <name>
- Example:
 - **create role** instructor
- Once a role is created we can assign “users” to the role using:
 - **grant** <role> **to** <users>

Roles Example

- **create role** instructor;
- **grant** *instructor* **to** Amit;
- Privileges can be granted to roles:
 - **grant select on** *takes* **to** *instructor*;
- Roles can be granted to users, as well as to other roles
 - **create role** *teaching_assistant*
 - **grant** *teaching_assistant* **to** *instructor*;
 - *Instructor* inherits all privileges of *teaching_assistant*
- Chain of roles
 - **create role** *dean*;
 - **grant** *instructor* **to** *dean*;
 - **grant** *dean* **to** Satoshi;

Authorization on Views

- **create view** *geo_instructor* **as**
(**select** *
from *instructor*
where *dept_name* = 'Geology');
- **grant select on** *geo_instructor* **to** *geo_staff*
- Suppose that a *geo_staff* member issues
 - **select** *
from *geo_instructor*;
- What if
 - *geo_staff* does not have permissions on *instructor*?
 - Creator of view did not have some permissions on *instructor*?

Other Authorization Features

- **references** privilege to create foreign key
 - **grant reference** (*dept_name*) **on** *department* **to** Mariano;
 - Why is this required?
- transfer of privileges
 - **grant select on** *department* **to** Amit **with grant option**;
 - **revoke select on** *department* **from** Amit, Satoshi **cascade**;
 - **revoke select on** *department* **from** Amit, Satoshi **restrict**;
 - And more!

End of Chapter 4